# Data Curation Project Progress Report

**Team Members:**

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## Project feedback incorporation

In our project proposal, we received feedback to use a more reliable dataset. Previously, we were using Kaggle, which is an open-source dataset. Since most of the data is supplied by users via crowdsourcing, there could be a lot of data inconsistencies in using a dataset from Kaggle. The fundamental concerns regarding **data provenance and reliability** presented an unacceptable risk to the credibility of our results and project deliverables.

To preemptively mitigate these significant risks and align our project with the better standards of data integrity and transparency, we decided to pivot to a publicly documented and officially sourced dataset from the City Of Chicago Data Portal.

### New Dataset

The **311 Service Requests received by the City of Chicago** dataset is more reliable and trustworthy as it represents real world dataset and is generated and maintained by the **City of Chicago** as a public record of its municipal operations. This official, government provenance establishes a high level of **accountability, documentation, and reliability.** Even though we had to change our data source, we are still well on track in terms of the timeline of our project. The Chicago 311 dataset can be accessed via this link.

# Dataset Details

| Column Name | Description | Data Type |
|---|---|---|
| SR_NUMBER | | Text |
| SR_TYPE | | Text |
| SR_SHORT_CODE | An internal code corresponding to the Service Request Type. This code allows for searching and filtering more easily than using the full SR_TYPE value. | Text |
| CREATED_DEPARTMENT | The department, if any, that created the service request. | Text |
| OWNER_DEPARTMENT | The department with initial responsibility for the service request. | Text |
| STATUS | | Text |
| ORIGIN | How the request was opened. Some values, such as Generated In House and Mass Entry result from the City's own operations. | Text |
| CREATED_DATE | | Floating Timestamp |
| LAST_MODIFIED_DATE | | Floating Timestamp |
| CLOSED_DATE | | Floating Timestamp |
| STREET_ADDRESS | | Text |
| CITY | | Text |
| STATE | | Text |
| ZIP_CODE | | Text |
| STREET_NUMBER | | Text |
| STREET_DIRECTION | | Text |
| STREET_NAME | | Text |
| STREET_TYPE | | Text |
| DUPLICATE | Is this request a duplicate of another request? | Checkbox |
| LEGACY_RECORD | Did this request originate in the previous 311 system? | Checkbox |
| LEGACY_SR_NUMBER | If this request originated in the previous 311 system, the original Service Request Number. | Text |
| PARENT_SR_NUMBER | Parent Service Request of the record if applicable. If the current Service Request record has been identified as a duplicate request, the record will be created as a child of the original request. | Text |
| COMMUNITY_AREA | | Number |
| WARD | | Number |
| ELECTRICAL_DISTRICT | | Text |
| ELECTRICITY_GRID | | Text |
| POLICE_SECTOR | | Text |
| POLICE_DISTRICT | | Text |
| POLICE_BEAT | | Text |
| PRECINCT | | Text |
| SANITATION_DIVISION_DAYS | | Text |
| CREATED_HOUR | The hour of the day component of CREATED_DATE. | Number |
| CREATED_DAY_OF_WEEK | The day of the week component of CREATED_DATE. Sunday=1 | Number |
| CREATED_MONTH | The month component of CREATED_DATE | Number |
| X_COORDINATE | The x coordinate of the location in State Plane Illinois East NAD 1983 projection. | Number |
| Y_COORDINATE | The y coordinate of the location in State Plane Illinois East NAD 1983 projection. | Number |
| LATITUDE | The latitude of the location. | Number |
| LONGITUDE | The longitude of the location. | Number |
| LOCATION | The location in a format that allows for creation of maps and other geographic operations on this data portal. | Location |

## Use Case for New Dataset

Optimizing City Resource Allocation and Service Delivery

To analyze the entire service request lifecycle—from submission to completion and quantitatively assess the performance of different city departments and identify areas where resource optimization and process improvements are most needed.

## Project Status and Deliverables

| S.No | Tasks | Deadline | Responsibility | Status |
|---|---|---|---|---|
| 1 | Data Acquisition | Week1 | All Team Members | Completed |
| 2 | Ethical, legal, and policy constraints | Week 2 | Matthew Guan | Completed |
| 3 | Data Cleaning and Transformation | Week2 – Week3 | Murali Natarajan | Completed |
| 4 | Data Governance (De-Identification) | Week3 | Murali Natarajan | Completed |
| 5 | Data Analysis -Unsupervised Machine Algorithm | Week3 – Week 6 | Ramitha Kotarkonda | Completed |
| 6 | Data Analysis - determining feature importance via prediction models | Week4 - Week6 | Matthew Guan | Completed |
| 7 | Progress Report | Week6 | All Team Members | Completed |
| 8 | Managing Data Quality | Week3 - Week6 | Ramitha Kotarkonda | In Progress |
| 9 | Artifacts & Workflow | Week6 – Week7 | Murali Natarajan | In Progress |
| 10 | Reproducibility & Transparency | Week 7 | Murali Natarajan | |
| 11 | Metadata & Documentation | Week8 – Week9 | Ramitha Kotarkonda | |
| 12 | Final Report | Week 10 - 12 | All Team Members | |

We were able to complete all the milestones mentioned in the project plan up to October 27th. In week 2, we worked on identifying the ethical, legal, and policy constraints. Even though the 311 data is anonymized, the location fields such as latitude, longitude, and zip code can be linked back to an individual, which is a violation of privacy laws. We can use the dataset without legal constraints, because it is stated under the city of Chicago's terms of use. If we publish this data, then we will need to follow the steps listed in the terms and conditions and not misuse the data.

## Data Cleaning and Deidentification

Below is the Summary of Data Cleaning and Deidentification task performed on the Data.

# Data Cleaning Summary Report

### File Information

- **Input File:** `311_Service_Requests_20251022.csv`
- **Output File:** `311_Service_Requests_CLEANED_20251022.csv`
- **Script Run Date:** `2025-10-25 14:41:14`

### Initial Data State (Before Cleaning)

- **Initial Row Count:** 199,999
- **Initial Column Count:** 39
- **Duplicate Records (Flagged as DUPLICATE=True):** 5,690

### Cleaning Actions Performed

1. **Dropped Duplicate Records (Rows where DUPLICATE was True):** 5,690
2. **Dropped Unlocatable Records (Missing Address/Coords):** 205
3. **Imputations/Standardizations:** CITY/STATE standardized and imputed, CREATED_DEPARTMENT filled with 'Unknown', ZIP_CODE converted to string.
4. **Feature Engineering:** Calculated `RESOLUTION_TIME_HOURS` and split `CREATED_DATE` into `CREATED_DATE_PART` / `CREATED_TIME_PART` .

### Final Data State (After Cleaning)

- **Final Row Count:** 194,104
- **Final Column Count:** 36
- **Total Rows Dropped:** 5,895

## De-Identification (K-Anonymity) Summary Report

This report summarizes the de-identification process applied to the cleaned service request data to enforce **K-anonymity** and protect privacy.

---## K-Anonymity Configuration

- **K-Threshold:** $k = 5$
- **Quasi-Identifiers (QIAs) Used:** COMMUNITY_AREA, WARD, POLICE_DISTRICT, ZIP_CODE

---## Data Suppression and Retention

| Metric | Before De-identification | After K-Anonymity |
| --- | --- | --- |
| **Total Records** | 194,104 | 193,841 |
| **Records Suppressed** | N/A | 263 (Records with group size < $k$) |
| **Shape** | 194104 rows, 36 cols | 193841 rows, 28 cols |

---## Generalization and Masking Details The following columns were **generalized (masked)** to reduce their precision while retaining analytic utility:

| Column | Generalization Method |
| --- | --- |
| ZIP_CODE | 3-digit mask |
| LATITUDE | Rounded to 3 decimal places |
| LONGITUDE | Rounded to 3 decimal places |

### Columns Removed

The following columns were **removed entirely** as they were direct or high-precision identifiers:

- **Explicitly Dropped:** STREET_ADDRESS, STREET_NUMBER, STREET_NAME, STREET_DIRECTION, STREET_TYPE, LOCATION
- **Redundant Coordinates Dropped (after generalization):** X_COORDINATE, Y_COORDINATE

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## Data Analysis -Unsupervised Machine Algorithm

We worked on creating an unsupervised machine learning algorithm during weeks 3-6. We reduced dimensionality reduction using PCA. We used the explained variance ratio to manage quality and select the most optimal number of components, which was 3. Next, we moved on to creating a KMeans algorithm. We used the inertias to determine the optimal number of clusters and to ensure we are maintaining data quality. The chart in segmentation_model.ipynb shows us that 4 is the optimal number of clusters. Next, we used the clusters that were outputted by the cluster algorithm to understand what each of the clusters means, our key takeaways and recommendations.

### Determining feature importance via prediction models

We worked on determining feature importance via prediction models. Our goal was to determine the features that were most important in predicting the time it takes for a 311 request from Chicago to be resolved. First, I created a new derived feature from the dataset called 'Time to Close'. The value of this new feature is the difference between the close time and the creation time. Next, we trained a Random Forest regression model on the data. We decided on Random Forest because the predictions from many trees are averaged together, which reduces overfitting yet helps capture complex relationships among the data. Finally, we evaluated model performance via several metrics: mean squared error, mean absolute error, and $R^2$.

## New challenges identified and scope adjustments

1. **Identify a New Dataset**

   Based in the feedback of the Original Dataset that was chosen we decided to llok for alternate dataset which from a trustworthy and reliable source .

   We were able to identify a new dataset from "City of Chicago" that we as a team felt would be better fit for project.

2. **Data Volume**

   The new dataset, being a comprehensive log of all 311 interactions since late 2018, presents a substantially larger data volume than the original selectio.

**Implement a Sampling Strategy:** We will adjust the scope to initially work with a **stratified sample** of the data.

3. **Data Quality**

   While the new dataset is officially sourced, it contains specific quality issues inherent to human-entered operational data.

   The data cleaning phase will be expanded to include specific steps to removing invalid entries.

4. **Challenges with the model**

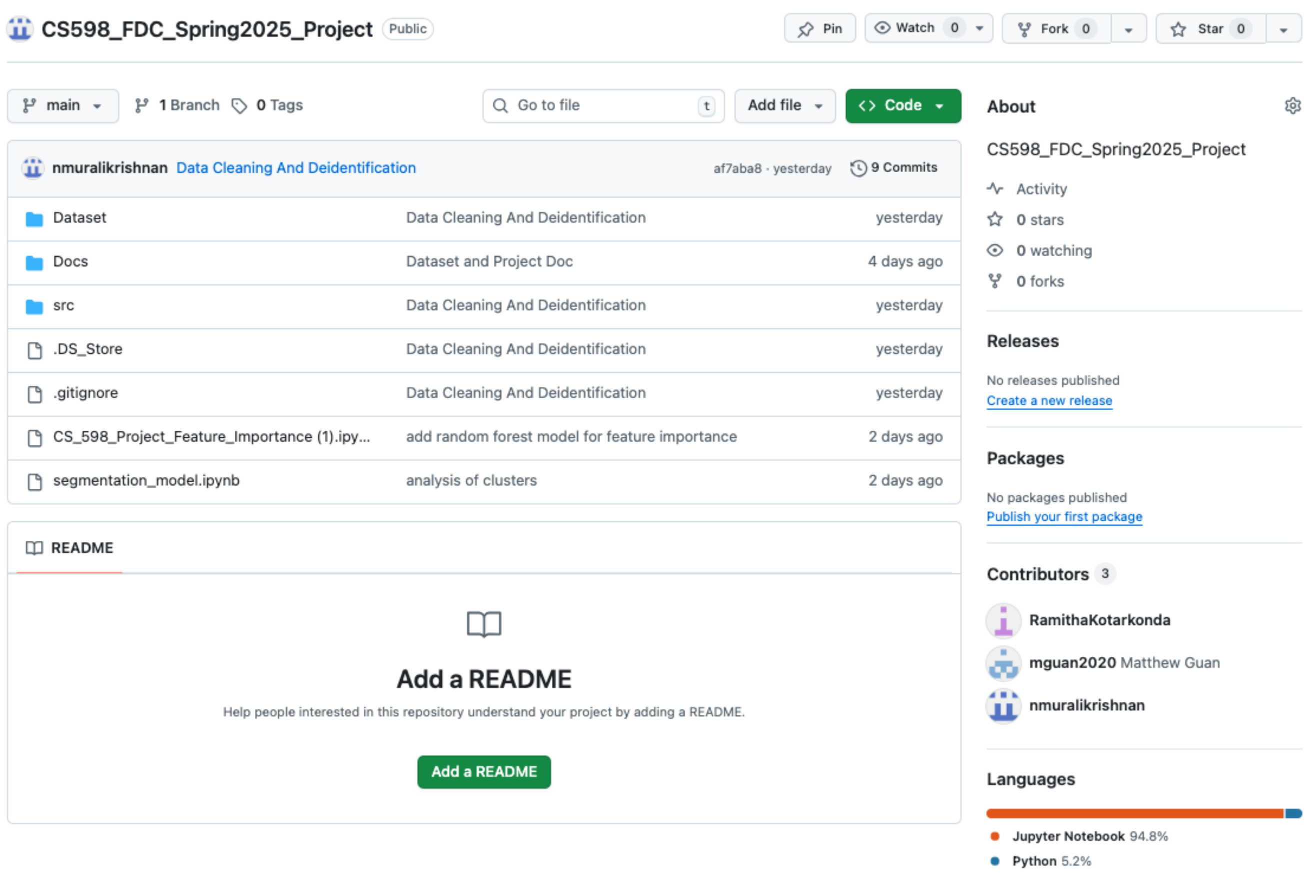
   Training the prediction model took longer than expected. There were many outliers in the data, since most calls were resolved in minutes, but a few took several days.

   To solve this issue, we applied log transform to reduce the influence of large values. We also plan to investigate methods to further improve accuracy of the predictions.

# Project Progress Artifacts:

This is our link to the github: [Activity · nmuralikrishnan/CS598_FDC_Spring2025_Project].

Just in case you can't access the link above, here is the proof of progress:



## Commits

main

All users | All time

Commits on Oct 25, 2025

- **Data Cleaning And Deidentification** — nmuralikrishnan committed yesterday — af7aba8
- **add random forest model for feature importance** — mguan2020 committed 2 days ago — 55dcab7

Commits on Oct 24, 2025

- **Read dataset for feature importance analysis** — mguan2020 authored 2 days ago — Verified — db26274
- **analysis of clusters** — RamithaKotarkonda committed 2 days ago — f5a7424
- **kmeans algorithm analysis** — RamithaKotarkonda committed 2 days ago — ebb1a58
- **kmeans algorithm** — RamithaKotarkonda committed 2 days ago — 421ee7d
- **encode, scale features, and dim reduction** — RamithaKotarkonda committed 2 days ago — bbd8ff0
- **read files and created a dataset with features** — RamithaKotarkonda committed 2 days ago — 3e764fe

Commits on Oct 23, 2025

- **Dataset and Project Doc** — nmuralikrishnan committed 4 days ago — beaf6f0